

Week 4: Sequences & Series

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※ Review

In general, a sequence is an ordered set of numbers, like $a_1, a_2, a_3, \dots, a_n$.

※ Arithmetic Sequences

An arithmetic sequence is one in which the difference between consecutive terms is constant. That is, the sequence $a_1, a_2, \dots, a_n$ is arithmetic when

$$a_2 - a_1 = a_3 - a_2 = \dots = a_n - a_{n-1}.$$

We often denote this common difference as d . Since every term is the same distance from the term before it, every term can be written as the first term, plus a multiple of the common difference. In general, the n th term of an arithmetic sequence is:

$$a_n = a_1 + (n - 1)d$$

In general, every sequence has an associated series, which is the sum of the terms in the sequence. We often write a series using summation notation:

$$\sum_{n=1}^k a_n = a_1 + a_2 + \dots + a_k.$$

The Σ indicates to substitute each of the integers from the lower summation bound to the upper bound, for n , and add all of those together. In the example above, the lower summation bound is $n = 1$, and the upper bound is $n = k$.

In general, if an arithmetic sequence has terms $a_1, a_2, \dots, a_n$ and common difference d , then the sum of those terms can be computed through the following

- Pair each term $\{a_1, a_n\}, \{a_2, a_{n-1}\}$, and so on. If the number of terms is odd, then the middle term

$$a_{\left(\frac{n+1}{2}\right)}$$

is on its own. Otherwise, every term would have a pair.

- Neatly enough, each pair has a positive sum. For example,

$$a_1 + a_n = (a_1 - d) + (a_n + d) = a_1 + a_n.$$

Other than the middle term, of course, in the case of an odd number of terms.

- There are technically $\frac{n}{2}$ total pairs. I say technically because this is true when n is even. But, if n is odd, it is a bit subtle. If n is odd, then middle term is on its own. We have $\frac{n-1}{2}$ entire pairs, but then the other middle term is left by its self. Technically, this gives it a value of $\frac{1}{2}$ of a pair. So, together this is $\frac{n}{2}$ pairs.
- Each pair has a sum of $a_1 + a_n = a_1 + [a_1 + (n - 1)d] = 2a_1 + (n - 1)d$. So, this sums to:

$$n \cdot \frac{a_1 + a_n}{2} = n \cdot \frac{2a_1 + (n - 1)d}{2}.$$

※ Geometric Sequences

In algebra, a geometric sequence, sometimes called a geometric progression, is a sequence of numbers such that the ratio between any two consecutive terms is constant. This constant is called the common ratio of the sequence.

For example, 1, 2, 4, 8 is a geometric sequence with common ratio 2 and 100, -50, 25, -25/2 is a geometric sequence with common ratio -1/2; however, 1, 3, 9, -27 and -3, 1, 5, 9, ... are not geometric sequences, as the ratio between consecutive terms varies.

More formally, the sequence $a_1, a_2, \dots, a_n$ is a geometric progression if and only if $a_2/a_1 = a_3/a_2 = \dots = a_n/a_{n-1}$. A similar definition holds for infinite geometric sequences. It appears most frequently in its three-term form: namely, that constants a , b , and c are in geometric progression if and only if $b/a = c/b$.

- Because each term is a common multiple of the one before it, every term of a geometric sequence can be expressed as the product of the first term and a power of the common ratio. Let a_1 be the first term, a_n be the n th term, and r be the common ratio of any geometric sequence; then, $a_n = a_1 r^{n-1}$.
- A common lemma is that a sequence is in geometric progression if and only if a_n is the geometric mean of a_{n-1} and a_{n+1} for any consecutive terms a_{n-1}, a_n, a_{n+1} . In symbols, $a_n^2 = a_{n-1}a_{n+1}$. This is mostly used to perform substitutions, though it occasionally serves as a definition of geometric sequences.

Now, let's try and compute the sum of a geometric series. Let the geometric series have value S . Then

$$S = a_1 + a_1 r + a_1 r^2 + \dots + a_1 r^{n-1}.$$

Factoring out a_1 , multiplying both sides by $(r - 1)$, and using the difference of powers factorization yields

$$S(r - 1) = a_1(r - 1)(1 + r + r^2 + \dots + r^{n-1}) = a_1(r^n - 1).$$

Dividing both sides by $r - 1$ yields $S = \frac{a_1(r^n - 1)}{r - 1}$, as desired.

Hence, a finite geometric series with first term a_1 , common ratio r not equal to one, and n total terms has a total sum equal to

$$\frac{a_1(r^n - 1)}{r - 1}.$$

An infinite geometric series converges if and only if $|r| < 1$; if this condition is satisfied, the series converges to $\frac{a_1}{1-r}$.

The proof that the series convergence if and only if $|r| < 1$ is an easy application of the ratio test from calculus; thus, such a proof is beyond the scope of this article. If one assumes convergence, there is an elementary proof of the formula that uses telescoping. Using the terms defined above,

$$S = a_1 + a_1 r + a_1 r^2 + \dots$$

Multiplying both sides by r and adding a_1 , we find that

$$rS + a_1 = a_1 + r(a_1 + a_1 r + \dots) = a_1 + a_1 r + a_1 r^2 + \dots = S.$$

Thus, $rS + a_1 = S$, and so $S = \frac{a_1}{1-r}$.

※ Telescoping Sequences

※ Recursive Sequences

※ Practice Problems

1. How many non-similar triangles have angles whose degree measures are distinct positive integers in arithmetic progression?
2. A grocer makes a display of cans in which the top row has one can and each lower row has two more cans than the row above it. If the display contains 100 cans, how many rows does it contain?
3. Find the value of

$$a_2 + a_4 + a_6 + a_8 + \cdots + a_{98}$$

if $a_1, a_2, a_3, \dots$ is an arithmetic progression with common difference 1, and $a_1 + a_2 + a_3 + \cdots + a_{98} = 137$.